

WEEK 8 – POWER BI REPORT

STUDENT PERFORMANCE DASHBOARD

Submitted By: Maheswari C

Roll No: E25DA049

Department: B.Sc Data Science and Artificial Intelligence

1.INTRODUCTION:

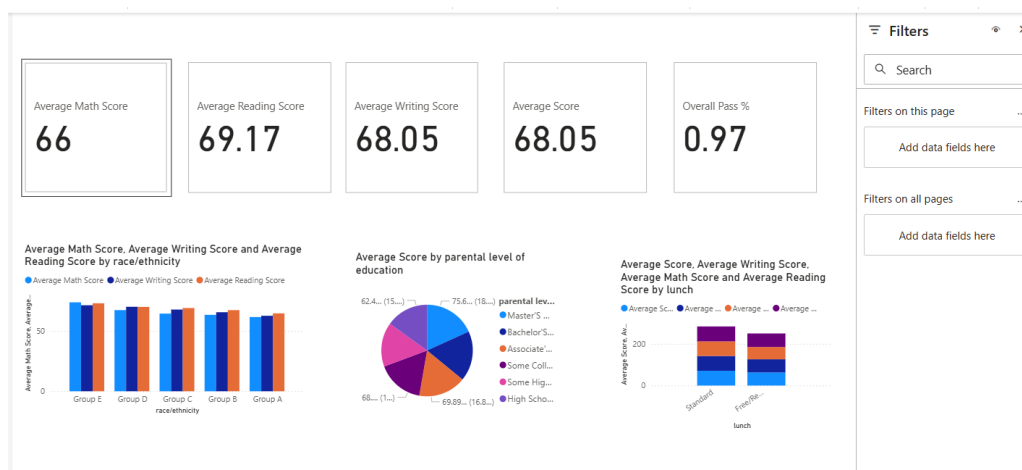
The Student Performance Dashboard was created using Microsoft Power BI to provide a clear and interactive view of students' academic performance. The dashboard focuses on subject-wise performance and comparison between different student categories.

2.DASHBOARD OVERVIEW

The dashboard contains KPI cards and interactive visualizations to analyze student performance.

The main KPIs displayed are:

- Average Math Score – 66
- Average Reading Score – 69.17
- Average Writing Score – 68.05
- Overall Average Score – 68.05
- Overall Pass Percentage – 97%



3.PERFORMANCE ANALYSIS

The dashboard compares student performance based on Race/Ethnicity, Parental Level of Education, Test Preparation Course, Lunch Type, Gender, and Subject.

Group E shows the highest average performance, while Group A shows the lowest average performance.

Students whose parents have a Master's degree show the highest average score, while the High School category shows the lowest average score.

4.TEST PREPARATION ANALYSIS

The dashboard compares students who completed the test preparation course with those who did not.

Students who completed the test preparation course show higher average scores than students who did not complete it.

5.LUNCH TYPE ANALYSIS

The dashboard compares students based on lunch type.

Students with standard lunch show higher average performance compared with students receiving free or reduced lunch. The difference is more noticeable in Math performance.

6. SUBJECT ANALYSIS

The dashboard compares the performance of three subjects:

- Math
- Reading
- Writing

The subject comparison helps identify differences in students' performance across the three subjects.

7.INTERACTIVE FEATURES

The dashboard contains slicers for:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Subject

These slicers allow users to filter the dashboard and analyze specific student groups.

8.KEY INSIGHTS

- Overall student performance is represented through KPI cards.
- Group E has the highest average performance.

- Group A has the lowest average performance.
- Test preparation is associated with higher average scores in the dashboard.
- Standard lunch students show higher average performance.
- Subject-wise comparison provides a clear view of Math, Reading, and Writing performance.

9.CONCLUSION

The Student Performance Dashboard provides an interactive and easy-to-understand view of academic performance. The dashboard helps users compare student groups, subjects, and performance categories using KPI cards, charts, and slicers.

